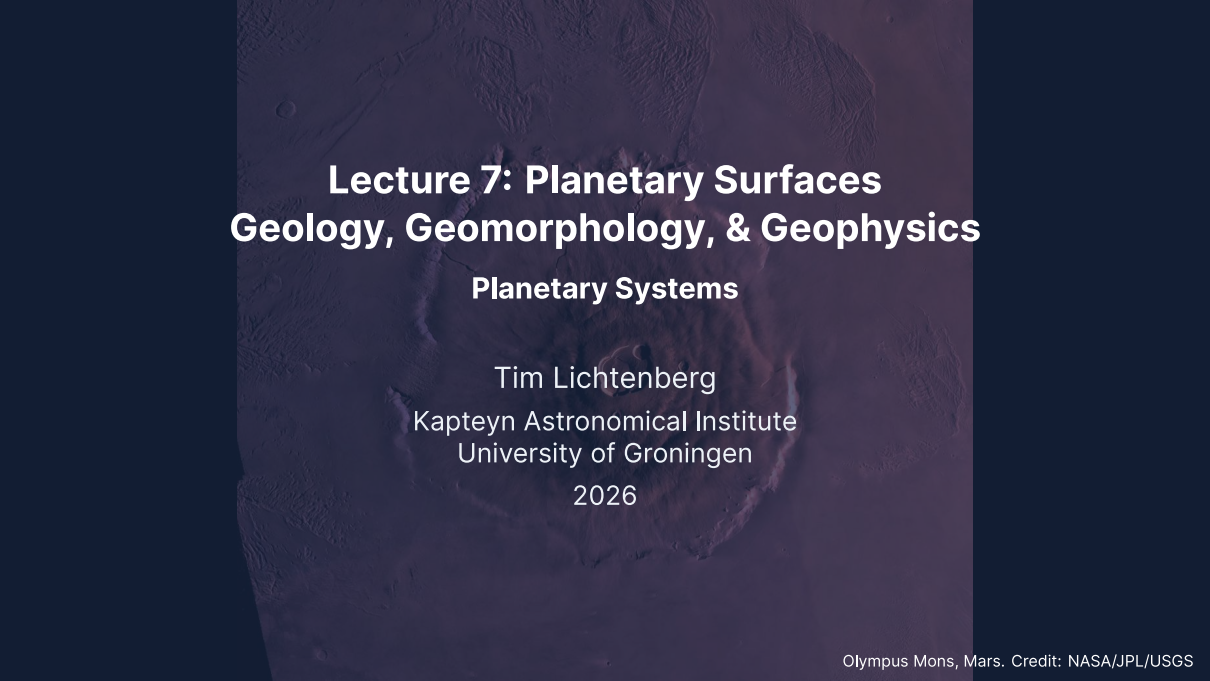


The background of the slide is a high-resolution, false-color image of the Martian surface, specifically focusing on the Olympus Mons region. The image shows the vast, dark, and heavily cratered plains of Mars, with the massive shield volcano of Olympus Mons dominating the center. The volcano's slopes are relatively smooth compared to the surrounding terrain, and its summit is visible as a bright, circular feature. The overall color palette is a mix of dark blues, purples, and greys, giving it a dramatic and scientific appearance.

Lecture 7: Planetary Surfaces Geology, Geomorphology, & Geophysics

Planetary Systems

Tim Lichtenberg
Kapteyn Astronomical Institute
University of Groningen

2026

Learning objectives

By the end of this lecture, you will be able to:

- Distinguish endogenic and exogenic surface processes
- Describe the three stages of impact crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ from dimensional analysis
- Use crater counting to determine relative and absolute surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system

Recap: Lectures 5 & 6

- Hydrostatic equilibrium and the scale height set an atmosphere's structure
- Greenhouse warming: $T_s > T_{\text{eff}}$; Clausius-Clapeyron sets where clouds form
- The carbonate-silicate cycle is Earth's long-term thermostat

Today: What shapes the *surfaces* of planets and moons?



1. Surface processes

Valles Marineris hemisphere of Mars, Viking Orbiter mosaic. Credit: NASA/JPL/USGS

Endogenic vs. exogenic processes

A surface is a geological record: processes that build landforms compete with processes that destroy them.

Endogenic (internal heat):

- Volcanism
- Tectonics (faulting, folding)

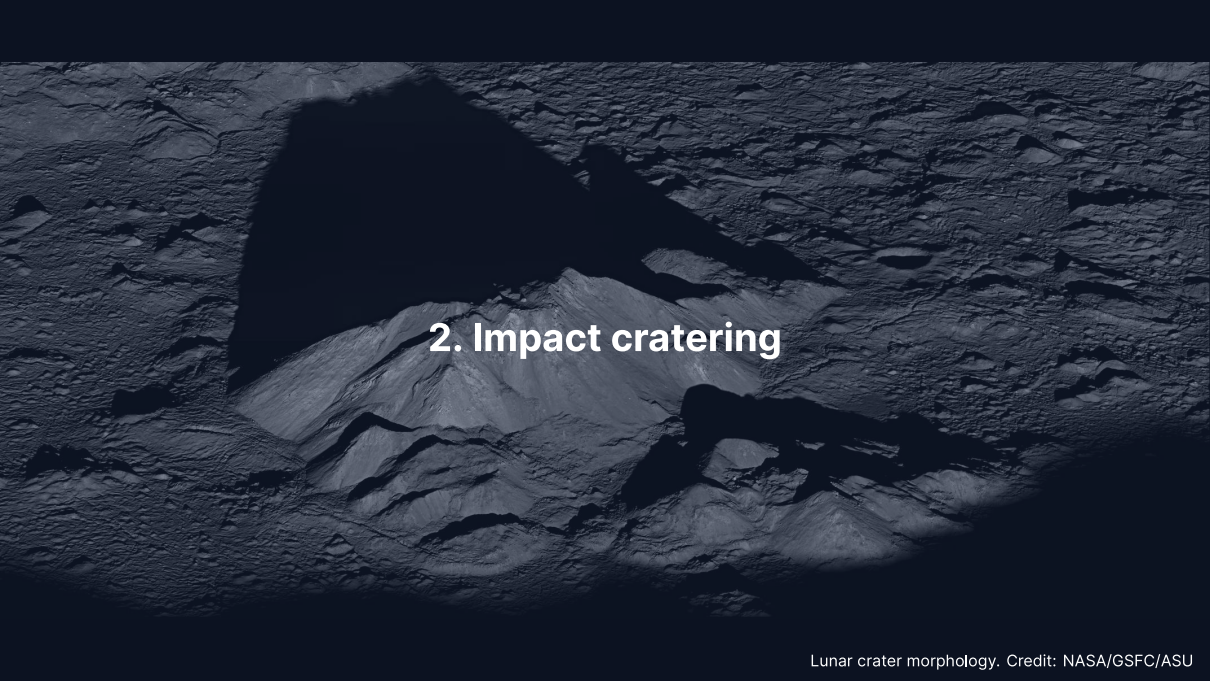
Exogenic (external forces):

- Impact cratering
- Erosion (wind, water, ice)

Hot interior: endogenic processes win. Dead interior: the surface keeps its ancient craters.

Dominant surface process by body

Body	Dominant process	Key evidence
Moon	Impact cratering	Saturated highland craters, >4 Gyr
Earth	Plate tectonics + erosion	Ocean floors <200 Myr; rapid weathering
Mars	Volcanism + impacts	Olympus Mons; cratered southern highlands
Venus	Volcanism	Volcanic plains cover ~80% of surface
Io	Tidal volcanism	~300–400 active volcanoes; age <1 Myr
Europa	Ice tectonics	Lineae, chaos terrain, very few craters

A black and white photograph of a lunar crater. The crater floor is covered in a dense field of smaller craters and rocks. A large, dark shadow is cast across the upper left portion of the image. In the center, a prominent, conical central peak rises from the floor. The text "2. Impact cratering" is overlaid in white on the central peak.

2. Impact cratering

The most universal geological process

- **Every** solid body in the solar system bears impact craters
- Without atmosphere or geology (Moon, Mercury), craters survive billions of years
- Impact velocities $10\text{--}70\text{ km s}^{-1}$, set by orbital mechanics (Lecture 2)

A 1 km rocky asteroid at 20 km s^{-1} carries $E_k = \frac{1}{2}mv^2 \approx 3 \times 10^{20}\text{ J}$, about 1000 times the largest nuclear weapon, released in under a second.

Three stages of crater formation

1. Contact and compression

- Shock wave into target and impactor
- ~100 GPa; melting and vaporisation
- Over in < 1 s

2. Excavation

- Rarefaction wave throws material out
- Transient cavity, depth:diameter $\approx 1:3$
- Ejecta blanket around the rim

3. Modification

- Cavity collapses under gravity
- Small: walls slump, **simple crater**
- Large: floor rebounds, **central peak**

3. Blackboard derivation: the crater scaling law

Caloris basin, Mercury. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

Setup: crater size from dimensional analysis

Goal: How does crater diameter D depend on impact energy E , target density ρ , and surface gravity g ?

In the **gravity regime** (craters $\gtrsim 100$ m, where gravity limits size):

$$D = C E^a \rho^b g^c$$

Dimensions (M, L, T):

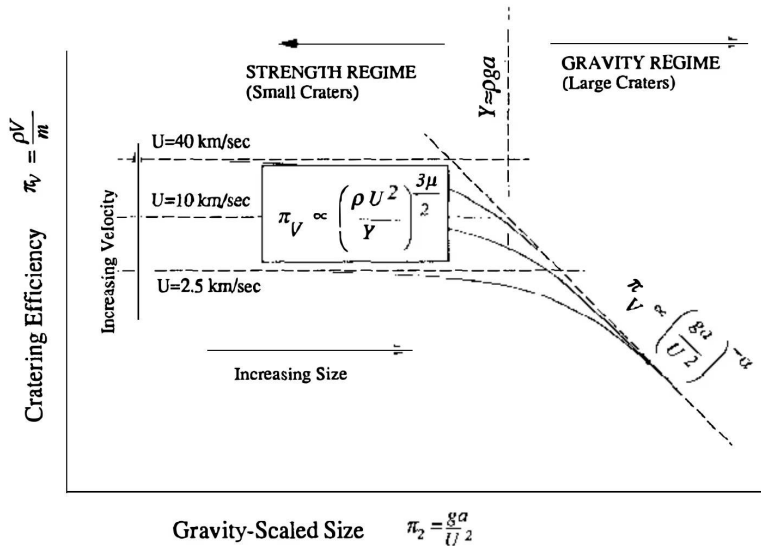
$$[D] = L$$

$$[E] = ML^2T^{-2}$$

$$[\rho] = ML^{-3}$$

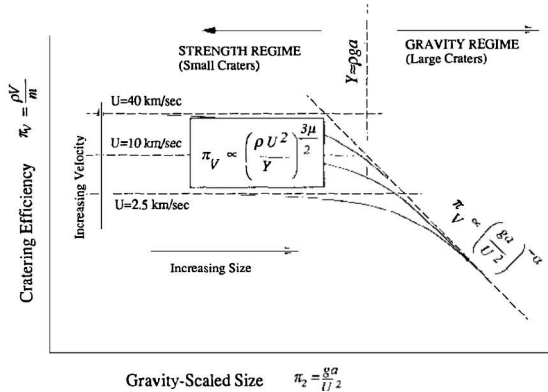
$$[g] = LT^{-2}$$

To the board.



The two cratering regimes for a target with strength, at impact velocities $U = 2.5, 10$ and 40 km s^{-1} . Reproduced from Holsapple (1993), Fig. 3.

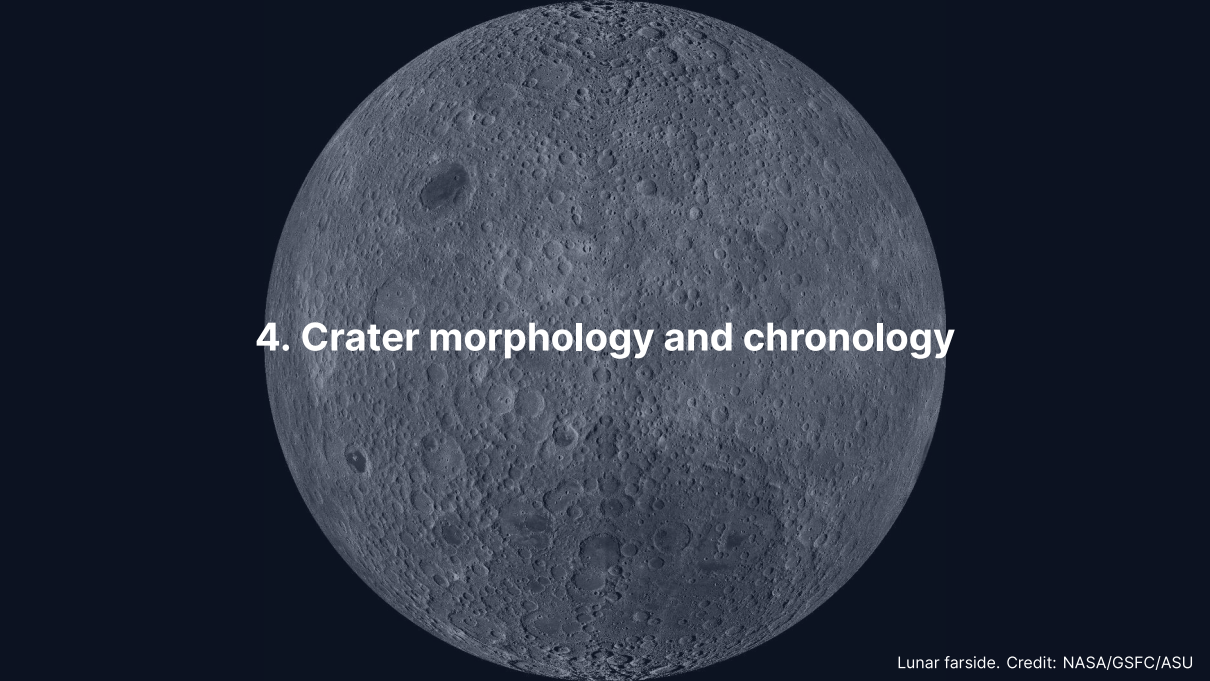
Where the fourth-root law applies



Credit: Holsapple (1993), Fig. 3

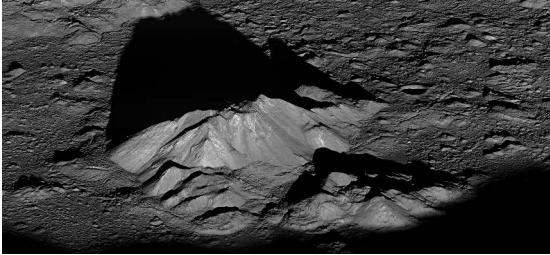
$D \sim (E/\rho g)^{1/4}$: doubling E grows D by only $2^{1/4} \approx 1.19$

- **Left, strength regime:** cohesion limits growth, curves separate
- **Right, gravity regime:** one law; planetary craters live here; below ~100 m on the Moon the law fails
- At fixed v and ρ : $E \propto L^3$, so $D \propto L^{3/4}$ (1 km impactor, 16.5 km crater)



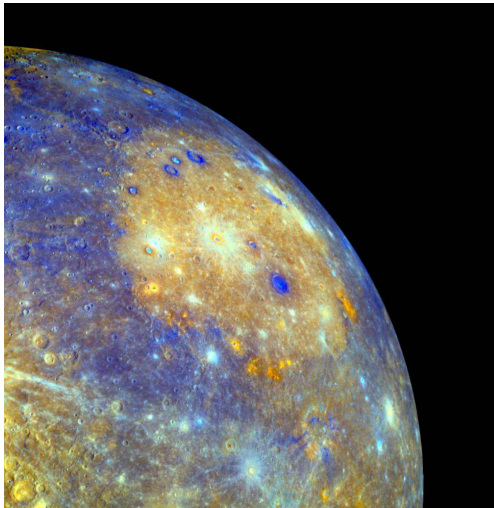
4. Crater morphology and chronology

Crater morphology: three classes



Credit: NASA/GSFC/ASU

1. **Simple** ($D < D_t$)
Bowl-shaped; depth:diameter $\approx$ 1:5
Moon: $D_t \approx 15$ km
2. **Complex** ($D_t < D \lesssim 300$ km)
Central peak, terraced walls, flat floor: gravity overcomes wall strength
3. **Multi-ring basins** ($D \gtrsim 300$ km)
Orientale (930 km), Caloris (1550 km), Hellas (2300 km)



Caloris basin, Mercury, ~1550 km in diameter: concentric rings, and volcanic plains (orange) that filled it after the impact. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

Transition diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\text{Moon}} \cdot \frac{g_{\text{Moon}}}{g}$$

Body	$g \text{ (m s}^{-2}\text{)}$	$D_t \text{ (km)}$
Moon	1.62	15
Mars	3.71	~6.5
Earth	9.81	~2.5
Mercury	3.70	~6.6

On Earth, virtually all craters larger than a few km are complex.

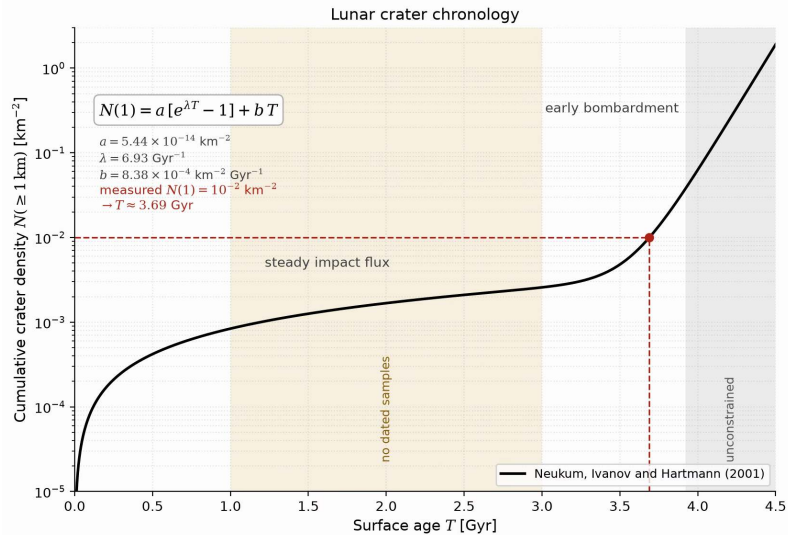
Crater counting: older surfaces carry more craters

The cumulative size-frequency distribution is a power law:

$$N(> D) = a D^{-b}, \quad b \approx 2-3$$

- $N(> D)$: craters per unit area larger than D ; a straight line on log-log axes
- Higher line = older surface: crater counting gives **relative** ages, until **saturation**, when new craters erase old ones
- Apollo and Luna samples date specific surfaces: the **absolute** calibration

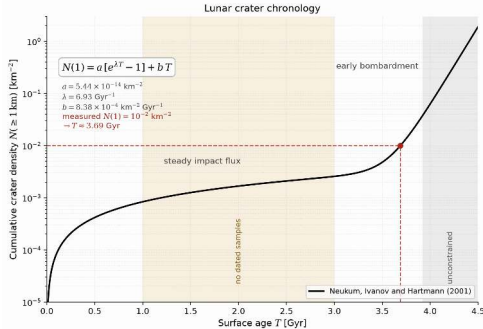
Source: Neukum et al. (2001), *Space Sci. Rev.*



The lunar chronology of Neukum, Ivanov and Hartmann (2001), with the intervals that no returned sample constrains. Course figure.

The lunar chronology

$$N(1) = a[e^{\lambda T} - 1] + b T$$



- Linear term: steady flux of the last ~3 Gyr; exponential term: the early bombardment before ~3.6 Gyr
- Read it backwards: $N(1) = 10^{-2} \text{ km}^{-2}$ gives ~3.7 Gyr
- Highlands saturated, >4 Gyr; maria 3.9–3.1 Gyr; extrapolated to Mars and Venus

The shaded bands hold no dated sample. There the curve is interpolation, not calibration.

Credit: Course figure, after Neukum, Ivanov and Hartmann (2001)



5. Volcanism

Olympus Mons, Mars. Credit: NASA/JPL/USGS

Effusive vs. explosive volcanism

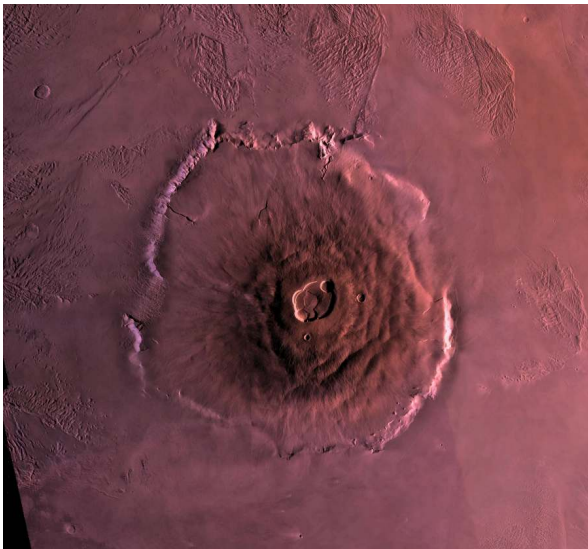
The key variable is **magma viscosity**, set by the SiO_2 content:

Basaltic, ~50% SiO_2 :

- Flows easily, volatiles escape
- **Effusive**: shields, flood basalts
- Moon, Mars, Io

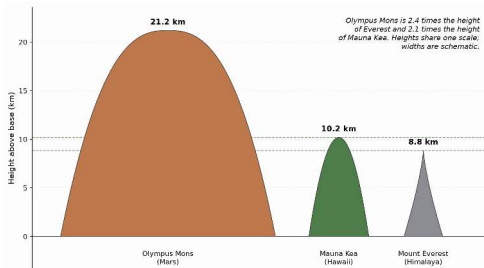
Silicic, >65% SiO_2 :

- Traps volatiles until failure
- **Explosive**: stratovolcanoes, ash
- Primarily Earth



Olympus Mons, Mars: a shield volcano ~600 km wide and ~21 km high, with nested summit calderas. Viking Orbiter mosaic. Credit: NASA/JPL/USGS.

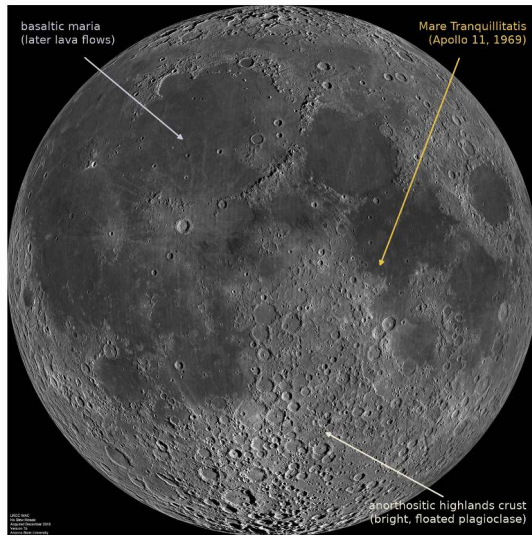
Why is Olympus Mons so large?



Credit: Course figure

Heights on one linear scale, widths schematic: Olympus Mons 21.2 km, Mauna Kea 10.2 km from the sea floor, Everest 8.8 km.

No plate motion: one plume feeds one edifice for billions of years.
Lower gravity, $g \approx 0.38 g_{\oplus}$, lets the crust hold a $\sim 2.6\times$ taller pile.



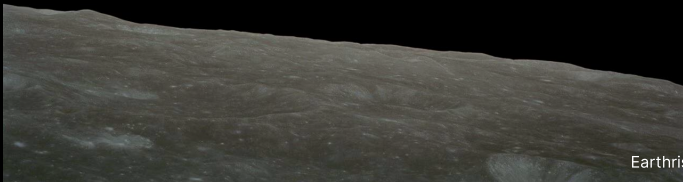
The lunar maria: flood basalts that filled the nearside basins between 3.9 and 3.1 Ga and cover ~16% of the surface. LROC wide-angle mosaic. Credit: NASA/GSFC/Arizona State University.

Volcanism across the solar system

Body	Style	Driving mechanism	Examples
Earth	Effusive + explosive	Internal heat + recycling	Hawaii, mid-ocean ridges
Moon	Flood basalts (extinct)	Residual heat (3.9–3.1 Ga)	Maria
Mars	Shield volcanoes	Mantle plumes, no plates	Olympus Mons, Tharsis
Venus	Lava plains + shields	Internal heat, stagnant lid	Maat Mons
Io	Ultramafic flows	Tidal heating	Loki Patera, Pele

Venus returns in Lecture 9, Io in Lecture 11.

Break

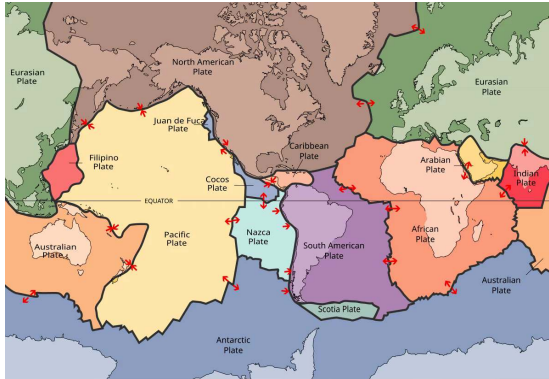


An aerial photograph of the Carrizo Plain in California, showing the San Andreas Fault as a prominent, winding linear depression. The landscape is arid and flat, with some small, dark, irregular features that might be water or wet mud. The fault line runs from the bottom center towards the top right, with a slight curve. The surrounding terrain is a mix of light and dark brown, indicating different soil types or vegetation. The overall tone is somewhat desaturated, giving it a historical or scientific feel.

6. Tectonics

The San Andreas Fault across the Carrizo Plain, California. Credit: Ikluft, Wikimedia Commons, CC BY-SA 4.0

Plate tectonics: Earth's unique regime



Credit: Wikimedia, CC BY-SA 3.0

Earth is the **only** body with active plate tectonics:

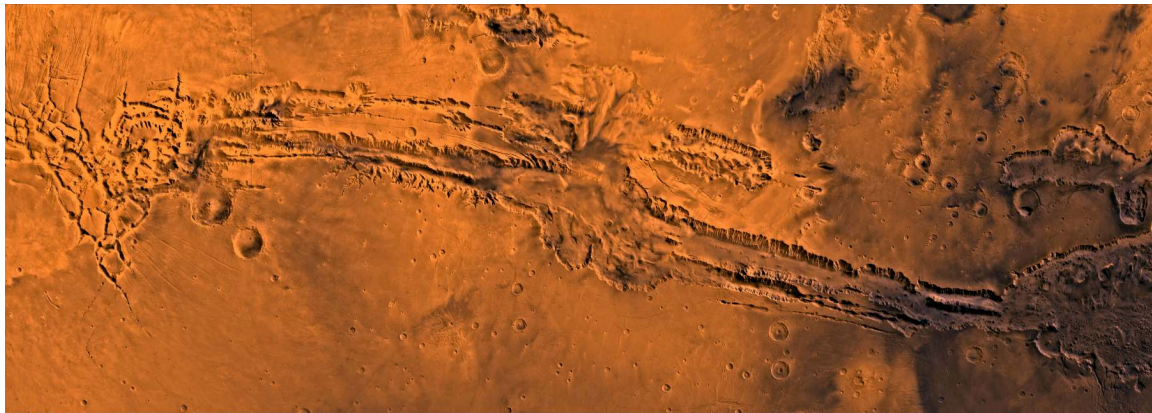
- ~15 rigid plates, moving at $1\text{--}10\text{ cm yr}^{-1}$
- **Divergent:** ridges, new crust
- **Convergent:** subduction, recycling
- **Transform:** horizontal sliding

Subduction closes the **carbonate-silicate cycle**.

Stagnant-lid convection

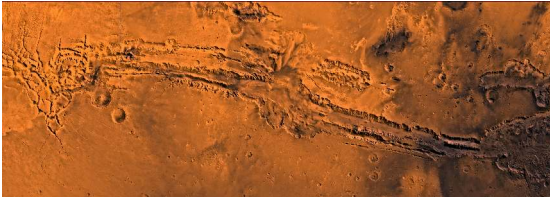
- Every other terrestrial body: one rigid lithospheric lid
- The mantle convects beneath it; heat leaves by conduction and occasional volcanism
- The thick lid insulates: the mantle stays warm, but melt reaches the surface only through rare plumes (Tharsis), and volcanism fades as the lid grows

Stagnant lid is the default for temperature-dependent viscosity. Breaking the lid, as Earth does, needs water weakening and self-sustaining damage.



Valles Marineris in a Viking Orbiter mosaic: Noctis Labyrinthus on the left, then the Melas, Candor and Coprates chasmata, and the chaotic terrain feeding Chryse Planitia on the right.
Credit: NASA/JPL/USGS.

Mars: Tharsis and Valles Marineris

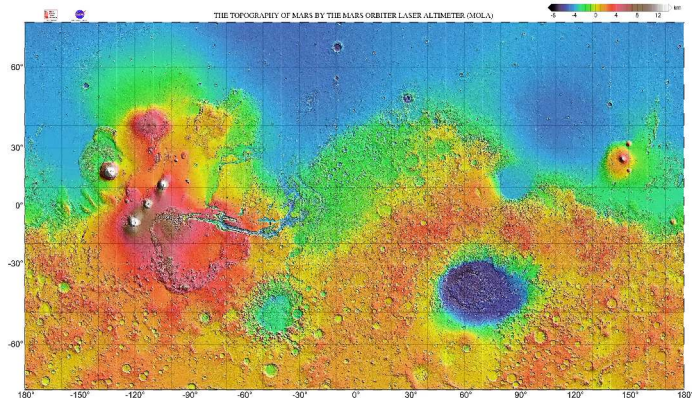


Credit: NASA/JPL/USGS

- **Tharsis:** volcanic plateau
~5000 km wide, ~10 km high,
above a mantle plume
- **Valles Marineris:** ~4000 km
long, up to 7 km deep, rifted
open by the Tharsis load

No plate boundaries, yet the
largest rift in the solar system.
One volcanic load did what
plate divergence does on Earth.

Mars topography: the hemispheric dichotomy



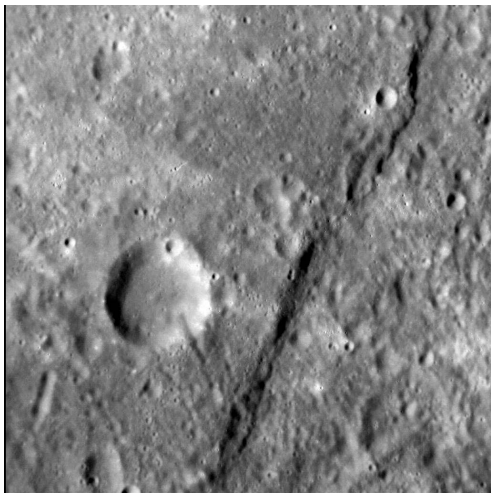
Credit: NASA/JPL/GSFC/MOLA Science Team

~6 km between the southern highlands and the northern lowlands. Tharsis with 4 giant volcanoes; Hellas at -8.2 km; Valles Marineris at the equator.



A lobate scarp on Mercury, a contraction thrust fault, imaged by MESSENGER. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

Mercury: a planet that shrank



Credit: NASA/Johns Hopkins APL/Carnegie Institution of

Washington

- The iron core cooled and solidified; the radius shrank by up to ~7 km
- The crust took up the contraction on **lobate scarps**: thrust faults hundreds of km long, 1–3 km high

The fault cuts craters, so the contraction is younger than they are. A cooling interior keeps deforming a stagnant lid.

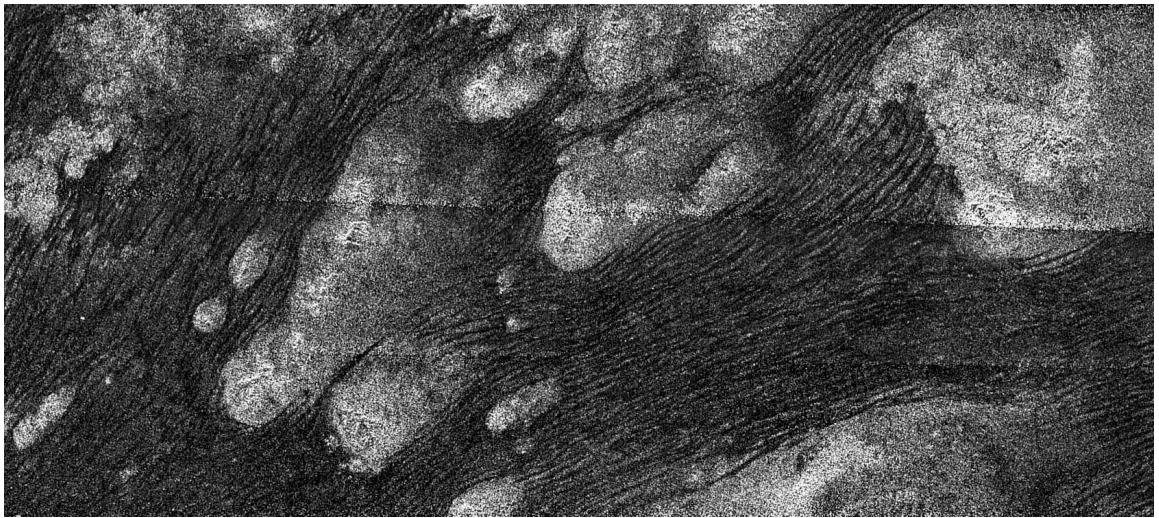


7. Erosion and weathering

Barchan dunes in Nili Patera, Mars, imaged by HiRISE. Credit: NASA/JPL-Caltech/University of Arizona

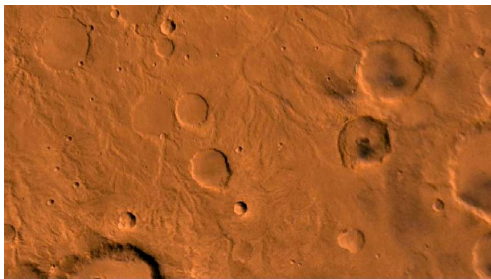


Barchan dunes in Nili Patera, Mars. Repeat HiRISE images show whole dunes advancing at sand fluxes comparable to Earth's, under an atmosphere 170 times thinner (Bridges et al. 2012). Credit: NASA/JPL-Caltech/University of Arizona.



Longitudinal dunes in Titan's Shangri-La region, Cassini SAR: organic sand, water-ice bedrock, the same aeolian physics. Credit: NASA/JPL-Caltech/ASI.

Fluvial (water) processes

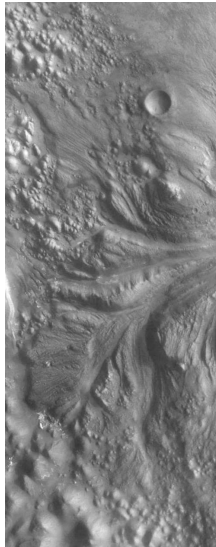


Credit: NASA/JPL/USGS

Mars valley networks: Noachian (>3.7 Ga), dendritic, sustained liquid water

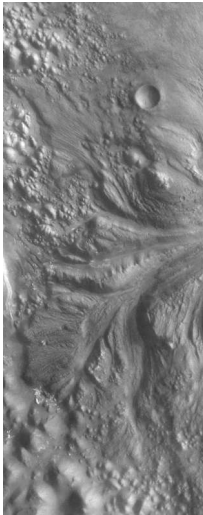
Outflow channels: hundreds of km long, catastrophic groundwater release

Titan: methane rivers; Huygens landed on rounded ice pebbles in a dry riverbed (2005)



A Martian outflow channel cutting through Aram Chaos. Credit: NASA/JPL-Caltech/MSSS.

Two kinds of water feature on Mars



- **Outflow channels** (here): braided islands, scoured troughs, one catastrophic flood
- **Valley networks**: dendritic, low discharge, long enough to need precipitation

Read the shape, not just the presence of water. One says flood, the other says climate.

Credit: NASA/JPL-Caltech/MSSS

Glacial and chemical weathering



Fedchenko Glacier, Pamir. Credit: NASA/ISS

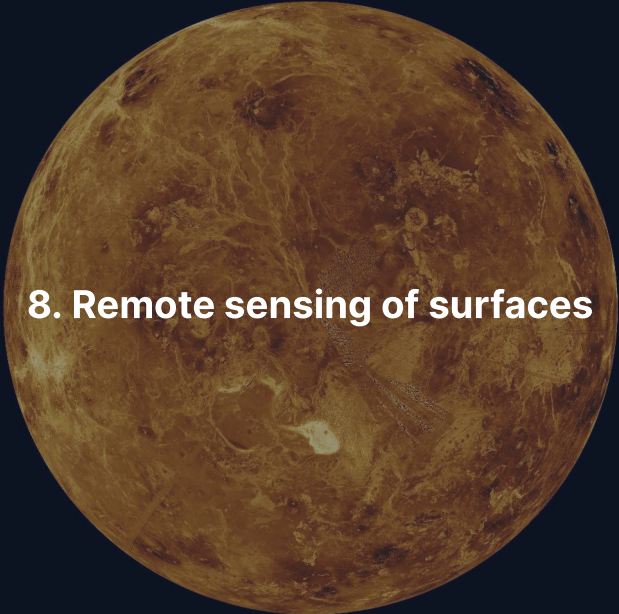


Clay-bearing unit, Glen Torridon, Gale crater. Credit:

NASA/JPL-Caltech/MSSS

Glacial: ice flows under its own weight. Earth's ice sheets covered ~30% of land at the Last Glacial Maximum; Mars has polar caps and buried mid-latitude ice.

Chemical: silicate weathering is Earth's carbon sink. On Mars, clays, sulfates and carbonates seen from orbit record past liquid water.



8. Remote sensing of surfaces

Venus in Magellan radar, the surface seen through an opaque atmosphere. Credit: NASA/JPL

Four ways to see a surface from orbit

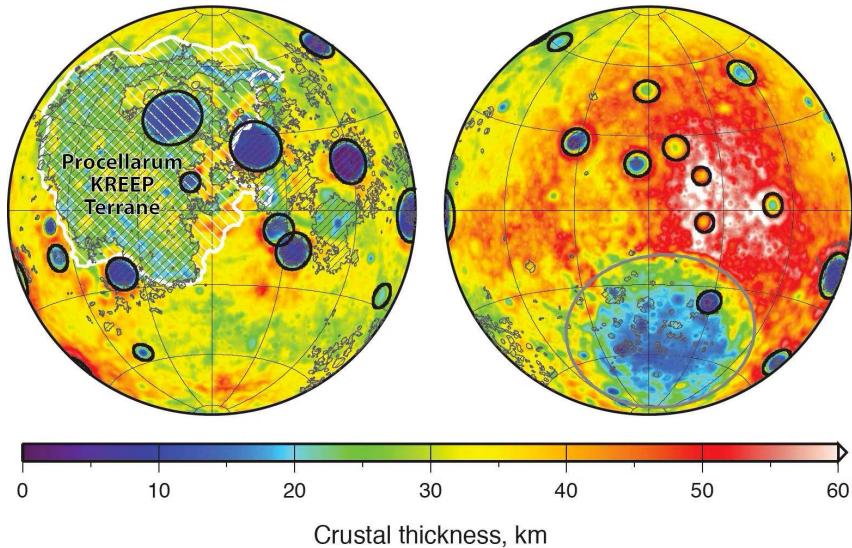
We land at a few sites. Almost all surface knowledge comes from orbit.

Reflectance spectroscopy minerals from absorption bands (CRISM at Mars: clays, carbonates)

Radar imaging through clouds and haze (Magellan at Venus, Cassini at Titan)

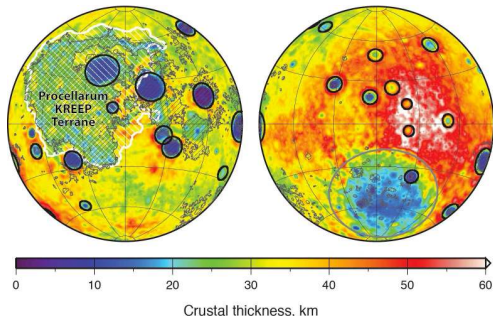
Laser altimetry topography to ~1 m (MOLA made the Mars map you just saw)

Gravity field mapping buried mass and crustal thickness (GRAIL at the Moon)



Lunar crustal thickness from GRAIL, nearside (left) and farside (right). Credit: NASA/MIT/GSFC/JPL-Caltech.

Weighing the lunar crust



Credit: NASA/MIT/GSFC/JPL-Caltech

- Two satellites track their separation; gravity plus topography gives the crust-mantle depth
- Mean thickness ~34–43 km; thin under the big basins, thick on the farside

Gravity sees what images cannot: a basin is a hole punched through most of the crust.



9. Regolith and space weathering

Apollo 11 bootprint in the lunar regolith. Credit: NASA

A shattered skin on every airless world

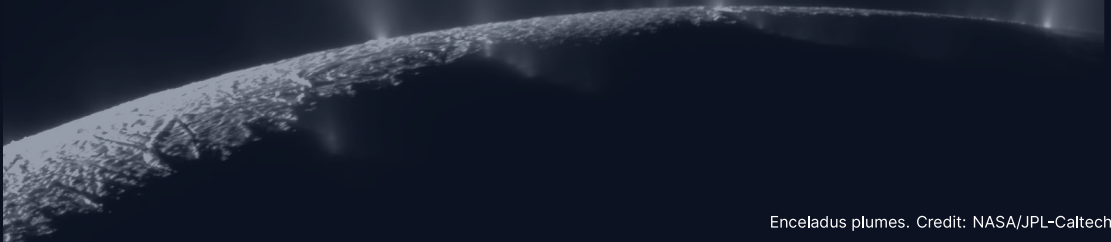


Credit: NASA/Apollo 11

Regolith: loose debris made by impact gardening; 2–4 m on the maria, 6–8 m on the highlands

Space weathering: solar wind and micrometeorites make nanophase iron; surfaces turn **darker and redder**, so fresh craters like Tycho look bright

10. Cryovolcanism on icy bodies



Cryovolcanism: volcanism on ice worlds

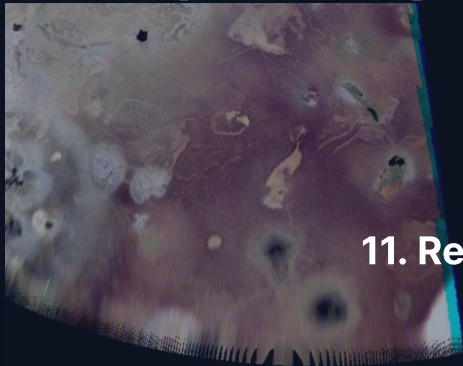
- The “magma” is water, ammonia-water or methane at 100–270 K; on Triton, nitrogen gas at ~38 K
- **Tidal heating** keeps oceans liquid under the ice shells
- Enceladus erupts water plumes, Europa resurfaces, Triton vents nitrogen

Liquid water far outside the habitable zone, kept by tides, not sunlight.
The bodies return in Lecture 11.



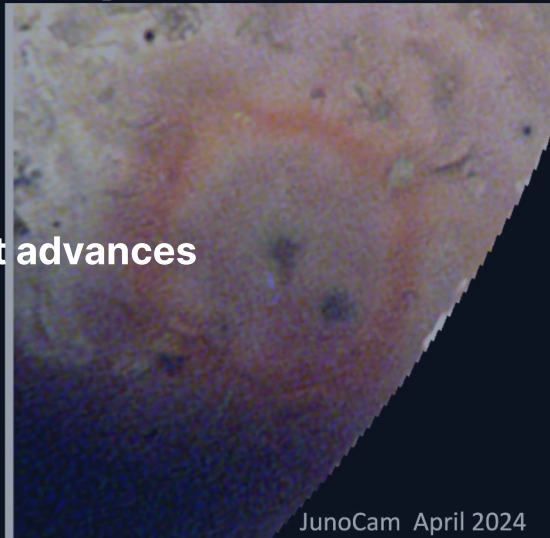
Double ridges and lineae near Pwyll crater on Europa, Galileo SSI. So few craters that the surface is only 40–90 Myr old. Credit: NASA/JPL-Caltech/University of Arizona/University of Colorado.

Change: New Bright Red Ring at Nusku



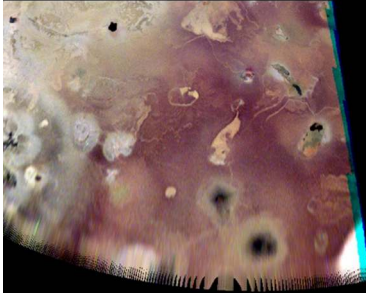
JunoCam February 2024

11. Recent advances

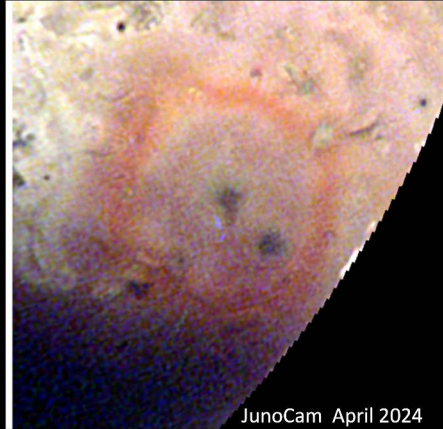


JunoCam April 2024

Change: New Bright Red Ring at Nusku



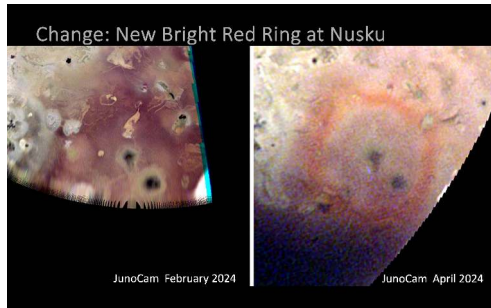
JunoCam February 2024



JunoCam April 2024

The Nuskū region on Io two months apart in 2024, JunoCam. Credit:
NASA/JPL-Caltech/SwRI/MSSS/Jason Perry.

Catching an eruption in the act on Io



Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry

- February 2024: nothing unusual around the vent
- April 2024: a red ring of fresh sulfur-rich pyroclastic deposits

Io is the one body beyond Earth where volcanic resurfacing shows up between two flybys. Juno gives the first sustained monitoring since Galileo.

Summary

1. Surfaces record **endogenic** (volcanism, tectonics) against **exogenic** (impacts, erosion) processes
2. Crater scaling law: $D \sim (E/\rho g)^{1/4}$, a fourth root of the energy
3. Crater counting plus Apollo calibration gives absolute surface ages
4. No plate tectonics: giant volcanoes, one rift, a shrinking Mercury; Earth alone breaks its lid
5. Fluvial features on Mars are strong evidence for past liquid water

Next: Lecture 8 goes underneath, to the interiors that drive the endogenic half of this record.

Questions?

Next: Lecture 8, Planetary interiors: structure, composition & dynamics

Thu 1 Oct, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>